

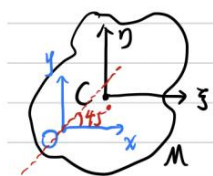
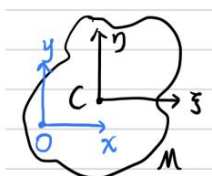
8、9、16 作业答案

8.2 质量为 M 的薄片,其质心为 $C, C\xi, C\eta$ 是它在薄片平面内的主轴。
 O 点在薄片平面内, Ox, Oy 相应地与 $C\xi, C\eta$ 平行. 求证: 薄片相对于 Oxy 的
 惯性积是 Mab , 其中 a, b 是 C 点在 Oxy 中的坐标。

如果薄片关于 $C\xi, C\eta$ 的主惯量分别是 nMa^2, mMb^2 , 并且薄片在 O 点有
 一与 Ox 轴成 45° 角的主轴, 求证

$$(n-1)a^2 = (m-1)b^2.$$

$$\begin{aligned} \because \forall m_i &= m_i(\xi, \eta), m_i(x, y) = C(x, y) + m_i(\xi, \eta) \\ \therefore \int xy \, dm &= \int (a+\xi)(b+\eta) \, dm \\ &= Mab + \underbrace{a \int \eta \, dm}_{\text{C是质心}} + b \int \xi \, dm + \int \xi \eta \, dm \\ &= Mab \end{aligned}$$

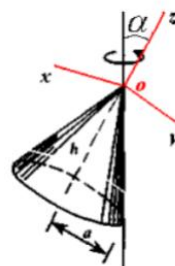


$$\begin{aligned} \therefore \int x^2 \, dm &= \int (a+\xi)^2 \, dm = a^2 M + \int \xi^2 \, dm \\ \int y^2 \, dm &= \int (b+\eta)^2 \, dm = b^2 M + \int \eta^2 \, dm \\ \therefore I_x &= b^2 M + nMa^2, \quad I_y = a^2 M + mMb^2 \\ \text{且 } \int xy \, dm &= Mab \\ \therefore I_x \cdot x'^2 + I_y \cdot y'^2 - 2I_{xy} \cdot x'y' &= k^2 \\ M(na^2+b^2)x'^2 + M(a^2+mb^2)y'^2 - 2Mab \cdot x'y' &= k^2 \\ \text{令 } \begin{bmatrix} x' \\ y' \end{bmatrix} &= \begin{bmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}, \text{ 则 } \begin{cases} x' = \frac{1}{\sqrt{2}}x - \frac{1}{\sqrt{2}}y \\ y' = \frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y \end{cases} \\ \therefore (na^2+b^2)(x'-y')^2 + (a^2+mb^2)(x'+y')^2 - 2ab(x'-y')(x'+y') &= k^2 \\ \therefore -(na^2+b^2) + (a^2+mb^2) &= 0 \\ \therefore (n-1)a^2 - (m-1)b^2 &= 0 \end{aligned}$$

我直接由椭圆对称性, $I_x = I_y$, 即得 $(n-1)a^2 = (m-1)b^2$

8.5 均质圆锥体高为 h , 底半径为 a , 质量为 m . 求对于一条母线的转动
 惯量。

$$\begin{aligned} I_x &= \int_M (y^2 + z^2) \, dm = \int_V (y^2 + z^2) \rho \, dV = \int_V (y^2 + z^2) \rho \, dx \, dy \, dz \\ \text{令 } x &= r \cos \theta, \quad y = r \sin \theta, \quad z = z, \quad \left| \frac{\partial(x, y, z)}{\partial(r, \theta, z)} \right| = r, \quad r \leq \frac{z}{h} a \\ \therefore I_x &= \int_{-h}^0 dz \int_0^{2\pi} d\theta \int_0^{\frac{za}{h}} (r^2 \sin^2 \theta + z^2) \rho \, r \, dr \\ &= \frac{\rho}{20} a^4 h \cdot \pi + \frac{\rho}{5} \pi a^2 h^3 \\ \int_0^{\frac{za}{h}} \sin^2 \theta \, d\theta &= \frac{\pi}{2} \times \frac{(a-h)}{\pi h}, \quad \forall \theta | z=0 \\ \therefore \rho &= \frac{m}{V} = \frac{3m}{\pi a^2 h} \\ \therefore I_x &= \frac{3}{20} ma^2 + \frac{3}{5} mh^2 = I_y \\ \text{且 } I_z &= \int_V (x^2 + y^2) \rho \, dx \, dy \, dz = \int_{-h}^0 dz \int_0^{2\pi} d\theta \int_0^{\frac{za}{h}} r^2 \rho \, r \, dr \\ &= \frac{\pi}{10} a^4 h \cdot \rho = \frac{3}{10} ma^2 \\ \therefore \vec{O\bar{S}} &= (0, \frac{a}{\sqrt{a^2+h^2}}, \frac{h}{\sqrt{a^2+h^2}}) \\ \therefore I_S &= (\frac{a}{\sqrt{a^2+h^2}})^2 I_y + (\frac{h}{\sqrt{a^2+h^2}})^2 I_z = \frac{ma^2}{20} \times \frac{3a^2+8h^2}{a^2+h^2} \end{aligned}$$



8.6 解 建立图示 cXY 坐标系,

可以证明 X, Y 是惯性主轴, 且

$$J_X = J_Y$$

$\therefore y$ 轴是对称轴,

$\therefore J_{XY} = J_{ZY} = 0$, 又因为

xy 平面对称面, 所以

$$J_{XZ} = J_{YZ} = 0, \quad XY \text{ 轴是}$$

主轴。再证明 $J_X = J_Y$

设 J_Y 已求得, 则由对称性知 $J_\xi = J_Y$, 在 cXY 平面内,

$$J_Y = J_\xi = J_X \cos^2 \alpha + J_Y \sin^2 \alpha = J_X \cos^2 \alpha + J_Y \sin^2 \alpha$$

$$J_Y (1 - \sin^2 \alpha) = J_X \cos^2 \alpha \quad \Rightarrow \quad J_X = J_Y$$

由此可见, 求得 J_Y 就是过质心任一轴的转动惯量。

计算正 n 边形内任一三角形 ACB 的 J_x, J_y

$$J_x = \int_0^{R \cos \alpha} y^2 \rho 2x dy = \frac{m}{2n} R^2 \cos^2 \left(\frac{\pi}{n} \right)$$

$$J_y = \int_0^{R \cos \alpha} \frac{1}{12} \rho 2x (2x)^2 dy = \frac{m}{6n} R^2 \sin^2 \left(\frac{\pi}{n} \right)$$

由平面薄板 $J_z = J_x + J_y$, 正 n 边形对过质心 C 的整体 J_z 有

$$J_z = \sum J_z = n(J_x + J_y)$$

$$J_z = J_x + J_y = 2J_y = n(J_x + J_y)$$

$$J_y = \frac{n}{2} (J_x + J_y) = \frac{n}{2} \left(\frac{m}{2n} R^2 \cos^2 \left(\frac{\pi}{n} \right) + \frac{m}{6n} R^2 \sin^2 \left(\frac{\pi}{n} \right) \right)$$

$$\text{即 } J_y = \frac{m}{12} R^2 (2 + \cos \frac{2\pi}{n})$$

8.9 解 在主坐标下表示角速度, 角加速度, 动量矩和动能:

$$\vec{\omega} = \omega_x \vec{i} + \omega_y \vec{j} + \omega_z \vec{k}$$

$$\vec{\varepsilon} = \dot{\omega}_x \vec{i} + \dot{\omega}_y \vec{j} + \dot{\omega}_z \vec{k}$$

$$\vec{H}_0 = J_x \omega_x \vec{i} + J_y \omega_y \vec{j} + J_z \omega_z \vec{k}$$

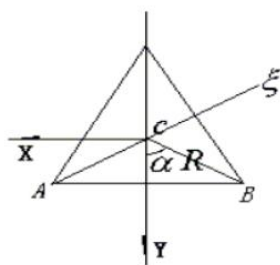
$$T = \frac{1}{2} (J_x \omega_x^2 + J_y \omega_y^2 + J_z \omega_z^2)$$

动量矩矢量和角加速度矢量垂直有:

$$\vec{H}_0 \cdot \vec{\varepsilon} = 0 = J_x \omega_x \dot{\omega}_x + J_y \omega_y \dot{\omega}_y + J_z \omega_z \dot{\omega}_z$$

$$\text{动能的变化率: } \frac{dT}{dt} = J_x \omega_x \dot{\omega}_x + J_y \omega_y \dot{\omega}_y + J_z \omega_z \dot{\omega}_z = 0$$

即动能 T 为常数

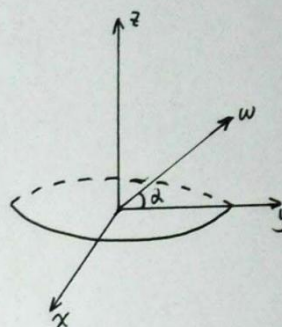


8.11. 解. 如右图所示, 有

$$\omega_y = \omega \cos \alpha \quad \omega_z = \omega \sin \alpha \quad \omega_x = 0$$

又因为角速度与进动角 ψ , 章动角 θ , 自转角 φ 的关系为

$$\begin{cases} \omega_x = \dot{\psi} \sin \theta \sin \varphi + \dot{\theta} \cos \varphi = 0 \\ \omega_y = \dot{\psi} \sin \theta \cos \varphi - \dot{\theta} \sin \varphi = \omega \cos \alpha \\ \omega_z = \dot{\psi} \cos \theta + \dot{\varphi} = \omega \sin \alpha \end{cases}$$



又由于不受外力矩作用, 则 $\omega_x, \omega_y, \omega_z$ 均为常值.

$$\text{从而可求得 } \theta = \frac{\pi}{2} - \alpha, \quad \varphi = 0 \quad \omega = \dot{\psi} = \omega$$

8.12 解由欧拉动力学方程:

$$I_1 \dot{\omega}_\xi + (I_3 - I_1) \omega_\eta \omega_\zeta = -\lambda I_3 \omega_\xi \quad (1)$$

$$I_2 \dot{\omega}_\eta + (I_1 - I_3) \omega_\xi \omega_\zeta = -\lambda I_3 \omega_\eta \quad (2)$$

$$I_3 \dot{\omega}_\zeta = -\lambda I_3 \omega_\zeta \quad (3)$$

由(3)得: $\frac{d\omega_\zeta}{\omega_\zeta} = -\lambda dt$ 且 $t=0$ 时, $\omega_\zeta = \omega_{\zeta 0}$

$$\therefore \omega_\zeta = \omega_{\zeta 0} e^{-\lambda t} \quad (4)$$

将(4)代入(1)(2)得:

$$\frac{d\omega_\xi}{dt} = \frac{I_1 - I_3}{I_1} \omega_\eta \omega_{\zeta 0} e^{-\lambda t} - \lambda \frac{I_3}{I_1} \omega_\xi \quad (5)$$

$$\frac{d\omega_\eta}{dt} = -\frac{I_1 - I_3}{I_1} \omega_\xi \omega_{\zeta 0} e^{-\lambda t} - \lambda \frac{I_3}{I_1} \omega_\eta \quad (6)$$

$$\frac{(5)}{\omega_\xi} + \frac{(6)}{\omega_\eta} \Rightarrow \omega_\xi \frac{d\omega_\xi}{dt} + \omega_\eta \frac{d\omega_\eta}{dt} = -\lambda \frac{I_3}{I_1} (\omega_\xi^2 + \omega_\eta^2)$$

$$\text{即 } d(\omega_\xi^2 + \omega_\eta^2) = -2\lambda \frac{I_3}{I_1} (\omega_\xi^2 + \omega_\eta^2) dt$$

$$\therefore \omega_\xi^2 + \omega_\eta^2 = A e^{-\frac{2\lambda I_3}{I_1} t} \quad A \text{ 为常数}$$

$$\text{设 } \omega_\xi = \sqrt{A} e^{-\frac{\lambda I_3}{I_1} t} \sin \varphi(t) \quad (7)$$

$$\omega_\eta = \sqrt{A} e^{-\frac{\lambda I_3}{I_1} t} \cos \varphi(t)$$

(7) 代入(5)式化简得:

$$\dot{\varphi} = \frac{I_3 - I_1}{I_1} \omega_{\zeta 0} e^{-\lambda t}$$

$$\text{令 } n = \frac{I_3 - I_1}{I_1} \omega_{\zeta 0}$$

$$\therefore \varphi = -\frac{I_3 - I_1}{\lambda I_1} \omega_{\zeta 0} e^{-\lambda t} + \varepsilon$$

$$\text{则 } \varphi = \frac{n}{\lambda} e^{-\lambda t} + \varepsilon \quad (\varepsilon \text{ 为常数}) \text{ 记 } a = \sqrt{A}$$

$$\omega_\xi = a \exp\left(-\frac{\lambda I_3}{I_1} t\right) \sin\left(\frac{n}{\lambda} e^{-\lambda t} + \varepsilon\right)$$

$$\text{即 } \omega_\eta = a \exp\left(-\frac{\lambda I_3}{I_1} t\right) \cos\left(\frac{n}{\lambda} e^{-\lambda t} + \varepsilon\right)$$

$$\omega_\zeta = \omega_{\zeta 0} e^{-\lambda t}$$

8.14. 解: (1) $\therefore \begin{cases} I_x \dot{\omega}_x + (I_z - I_y) \omega_y \omega_z = 0 \\ I_y \dot{\omega}_y + (I_x - I_z) \omega_z \omega_x = 0 \\ I_z \dot{\omega}_z + (I_y - I_x) \omega_x \omega_y = 0 \end{cases}$

将上面三式分别乘以 $\omega_x, \omega_y, \omega_z$, 然后相加起来再积分, 可知

$$I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2 = C \quad (1)$$

又 $\because \omega_x = \omega \cdot l_1, \quad \omega_y = \omega \cdot l_2, \quad \omega_z = \omega \cdot l_3$

l_1, l_2, l_3 均为常量, 代入(1)式, 有

$$(I_x l_1^2 + I_y l_2^2 + I_z l_3^2) \omega^2 = C$$

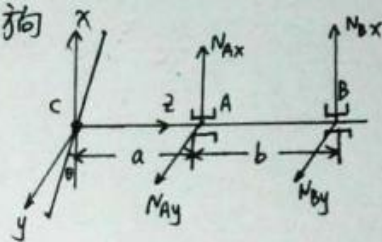
故 ω 为常量

(2) $\therefore \begin{cases} I_x \frac{d\omega_x}{dt} + (I_z - I_y) \omega_y \omega_z = L_x \\ I_y \frac{d\omega_y}{dt} + (I_x - I_z) \omega_z \omega_x = L_y \\ I_z \frac{d\omega_z}{dt} + (I_y - I_x) \omega_x \omega_y = L_z \end{cases}$

由于 $\dot{\omega}_x = \dot{\omega}_y = \dot{\omega}_z = 0$, 则

$$L = \sqrt{L_x^2 + L_y^2 + L_z^2} = \omega^2 \left\{ [(I_z - I_y) l_2 l_3]^2 + [(I_x - I_z) l_3 l_1]^2 + [(I_y - I_x) l_1 l_2]^2 \right\}^{1/2}.$$

8.16. 解: 如右图所示, 以质心 C 为原点, 建立一固结在螺旋桨上的坐标系 $Cxyz$ 令轴 Cz 与转轴重合, 正向为轴承 A 指向轴承 B , 轴 Cx 和 Cy 所在平面与轴 Cz 相垂直, A, B 两处的约束反力为 $N_{Ax}, N_{Ay}, N_{Bx}, N_{By}$, z 方向的约束反力为零



由质心运动定理, 可知

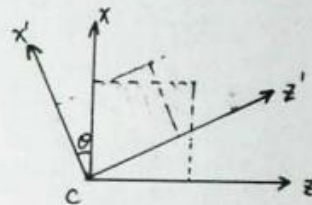
$$\begin{cases} -M(\varepsilon y_c + \omega^2 x_c) = N_{Ax} + N_{Bx} - Q \\ -M(\omega^2 y_c - \varepsilon x_c) = N_{Ay} + N_{By} \end{cases}$$

由动量矩定理, 可知

$$\begin{cases} I_{yz} \cdot \omega^2 - I_{xc} \cdot \varepsilon = -N_{Ay} \cdot a - N_{By} \cdot (a+b) \\ -I_{xz} \cdot \omega^2 - I_{yz} \cdot \varepsilon = N_{Ax} \cdot a + N_{Bx} \cdot (a+b) \end{cases}$$

由于 $\varepsilon = 0, x_c = y_c = 0, I_{yz} = 0$

$$\begin{aligned} \Rightarrow N_{Ax} &= Q - \frac{-I_{xz} \cdot \omega^2 \cdot a}{b} & N_{Bx} &= \frac{-I_{xz} \cdot \omega^2 - a \cdot a}{b} \\ &= Q + \frac{I_{xz} \cdot \omega^2 + a \cdot a}{b} \end{aligned}$$



$$\sum_{i=1}^n I_{xz} = \sum m_i x_i z_i$$

$$= \sum m_i (z'_i \sin \theta + x'_i \cos \theta) \cdot (z'_i \cos \theta - x'_i \sin \theta)$$

$$= \sum m_i x'_i z'_i \cos 2\theta + \sum m_i (z'^2_i - x'^2_i) \sin \theta \cos \theta$$

因为轴 Cx', Cz' 皆为惯性主轴, 则

$$\sum m_i x'_i z'_i \cos 2\theta = I_{x'z'} \cos 2\theta = 0$$

$$\therefore \underline{I_{xz}} = I_{xz} = \sum m_i (z'^2_i - x'^2_i) \sin \theta \cos \theta$$

$$= (I_{x'} - I_{z'}) \sin \theta \cos \theta$$

$$I_{x'} = 0 \quad I_{z'} = \frac{1}{12} m l^2 \quad \text{则} \quad I_{xz} = \frac{1}{12} m l^2 \sin \theta \cos \theta$$

$$= \frac{1}{12} \times 7.5 \times 1 \times \sin(0.015 \times 57.32) \cos(0.015 \times 57.32)$$

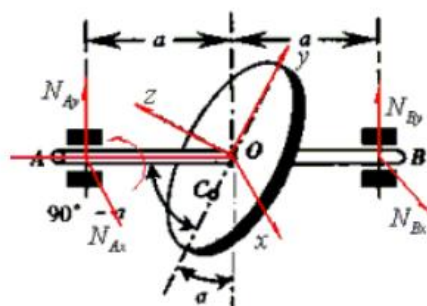
$$= 0.0094$$

$$\omega = 2\pi n = 100\pi \text{ rad/s}$$

$$\therefore N_{Ax} = Q + \frac{I_{xz} \cdot \omega^2 + a \cdot a}{b} = 75 + \frac{0.0094 \times (100\pi)^2 + 75 \times 0.1}{0.25} = 3812 \text{ N}$$

$$N_{Bx} = -3737 \text{ N}$$

8.21 解 建立图示随体坐标 $oxyz$ 坐标系, 重力引起的约束反力为非动约束反力, 只分析动约束反力且在随体标架中分解



$$N_{Ax}, N_{Bx}, N_{Ay}, N_{By}$$

x, y, z 都是惯性主轴:

$$I_z = \frac{1}{2}Mr^2 + Me^2, I_x = \frac{1}{4}Mr^2 + Me^2, I_y = \frac{1}{4}Mr^2$$

$$\omega_x = 0, \omega_y = -\omega \sin \alpha, \omega_z = \omega \cos \alpha \quad \text{且为常数}$$

代入欧拉动力学微分方程:

$$-(I_y - I_z)\omega_y\omega_z = (N_{By} - N_{Ay})a$$

$$0 = (N_{Ax} - N_{Bx})a \cos \alpha$$

$$0 = (N_{Ax} - N_{Bx})a \sin \alpha$$

由质心动量定理:

$$Me \sin \alpha \omega^2 = N_{Ay} + N_{By}$$

$$N_{Ax} + N_{Bx} = 0$$

可求得: $N_{Ax} = N_{Bx} = 0$

$$N_{By} = \frac{1}{2}Me \sin \alpha \omega^2 + \frac{\omega^2 \sin 2\alpha}{16a}M(r^2 + 4e^2)$$

$$N_{Ay} = \frac{1}{2}Me \sin \alpha \omega^2 - \frac{\omega^2 \sin 2\alpha}{16a}M(r^2 + 4e^2)$$

16.6 解: 对 M 有: 平面的约束反力 $N = Mg$

$$\text{则} \quad fMg = M\omega^2 R$$

$$\Rightarrow R = \frac{fg}{\omega^2}$$

物块能保持相对静止的位置在以转轴为圆

心, $\frac{fg}{\omega^2}$ 为半径的圆内

16.8 解 (1) 方法一: 由 $m\bar{a}_r = \bar{S}_e + \bar{S}_k + \bar{F}$
 $\bar{F} = \bar{0}$

将上式在 $\bar{v}_r$ 方向上投影得

$$m \frac{dv_r}{dt} = S_e \cos \theta \quad \because (\bar{S}_k \perp \bar{v}_r)$$

又 $\bar{v}_r = \dot{r}\bar{e}_r + r\dot{\varphi}\bar{e}_\varphi$

$$\therefore \cos \theta = \frac{\dot{r}}{v_r}$$

$$m \frac{dv_r}{dt} = \frac{\dot{r}}{v_r} \cdot mr\omega^2$$

积分得 $\frac{1}{2}v_r^2 = \frac{1}{2}\omega^2 r^2 + c$

$$v_r^2 - \omega^2 r^2 = \text{常量}$$

即 $\dot{x}^2 + \dot{y}^2 - \omega^2(x^2 + y^2) = \text{常量}$

(1) 方法二: 由科氏惯性力不作功, 牵连惯性力可看成有势力, 势函数为:

$$V = -\frac{1}{2}m\omega^2(x^2 + y^2)$$

非惯性系中机械能守恒:

$$\frac{1}{2}mv^2 - \frac{1}{2}m\omega^2(x^2 + y^2) = E$$

即 $\dot{x}^2 + \dot{y}^2 - \omega^2(x^2 + y^2) = \text{常量}$

(2) 方法一: 由非惯性系中的动量矩定理

$$\frac{d}{dt}(m\bar{r} \times \bar{v}_r) = \bar{r} \times \bar{S}_e + \bar{r} \times \bar{S}_k = \bar{r} \times \bar{S}_k$$

$$\bar{S}_k = -2m\omega\bar{v}_r$$

$$\bar{r} \times \bar{S}_k = 2m\omega r v_r \cos \theta \bar{k} = -2m\omega r \dot{r} \bar{k}$$

$$\therefore \frac{d}{dt}(\bar{r} \times \bar{v}_r) = -2\omega r \dot{r} \bar{k}$$

$$d(\bar{r} \times \bar{v}_r) = -2\omega r dr \bar{k}$$

$$\bar{r} = x\bar{i} + y\bar{j}; \quad \bar{v}_r = \dot{x}\bar{i} + \dot{y}\bar{j}$$

$$\therefore \bar{r} \times \bar{v}_r = -(y\dot{x} - x\dot{y})\bar{k}$$

$$\therefore d(y\dot{x} - x\dot{y} - \omega r^2) = 0$$

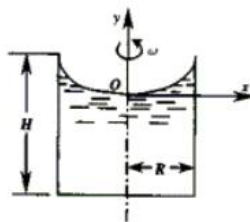
$$y\dot{x} - x\dot{y} - \omega(x^2 + y^2) = \text{常数}$$

(2) 方法二: 直角坐标直接投影方程积分.....

16.12 证明:

(1) 取动坐标系 oxy 为参考系, 取水面上—质点 m 分析

由于 $v_r = 0 \quad \therefore a_r = 0 \quad S_k = 0$



$$\therefore \quad o = \vec{S}_e + \vec{p} + m\vec{g}$$

(假设其他质点对其作用力垂直于表面)

于是有 $S_e \cos \theta = mg \sin \theta, \quad S_e = m\omega^2 x$

$$\therefore \quad \operatorname{tg} \theta = \frac{x\omega^2}{g} = \frac{dy}{dx}$$

积分得 $y = \frac{1}{2}x^2\omega^2/g + c$

解: $x=0$ 时 $y=0 \quad \therefore c=0$

$$y = \frac{1}{2}x^2\omega^2/g \text{ 为所求曲线}$$

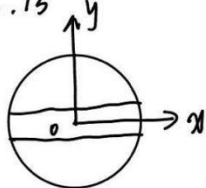
(2) 令 $x=R$, 则 $y = \frac{1}{2}R^2\omega^2/g$

设 注入液体最大高度 H' , 则

$$\pi R^2 H - \pi R^2 H' = \int_0^y dy \cdot \pi x^2$$

$$H' = H - \frac{R^2\omega^2}{4g}$$

16.13

建立沿弦位移 $x(t)$

$$\text{任一时刻 } F = m\omega^2 x(t)$$

$$\text{科氏力 } F_k = -2m\omega \cdot v$$

$$m\ddot{x} = m\omega^2 x(t)$$

$$\Rightarrow \ddot{x} = \omega^2 x(t)$$

$$x(0) = a, \quad \dot{x}(0) = 0$$

$$\text{求得 } x(t) = a \cdot \cosh(\omega t)$$

$$N - P - 2m\omega \dot{x} = 0$$

$$\dot{x} = a\omega \sinh(\omega t)$$

$$N = P + 2m\omega^2 a \sinh(\omega t)$$

$$= P + 2\frac{P}{g}\omega^2 a \sinh(\omega t)$$

16.12

$$S_k = 2m\omega v_r$$

$$S_c = m\omega^2 r$$

$$m\ddot{x}_r = m\omega^2 x_r$$

$$\Rightarrow \ddot{x}_r = \omega^2 x_r$$

$$x_r(0) = a, \quad \dot{x}_r(0) = v_{r1}$$

$$\Rightarrow x_r(t) = \frac{1}{2\omega} [(a\omega + v_{r1})e^{\omega t} + (a\omega - v_{r1})e^{-\omega t}]$$

$$v_r = \dot{x}_r = \frac{1}{2} [(a\omega + v_{r1})e^{\omega t} - (a\omega - v_{r1})e^{-\omega t}]$$

$$\text{由 } F_N - F_{1c} - F_{1e} \cos \theta = 0$$

$$\Rightarrow F_N = m\omega [(a\omega + v_{r1})e^{\omega t} - (a\omega - v_{r1})e^{-\omega t}] + m\omega^2 \sqrt{R^2 - a^2}$$

$$\text{令 } x_r(t) = l + a$$

$$\Rightarrow t = \frac{1}{\omega} \ln \frac{\omega(a+l) + v_{r2}}{a\omega + v_{r1}}$$

9.7 解: 由条件知 $v_1 = 200 \text{ km/h}$ $v_2 = 50 \text{ km/h}$

$$\therefore v_2^2 - v_1^2 = 2as$$

$$\Rightarrow a = 0.6 \text{ m/s}^2$$

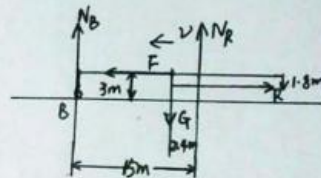
则惯性力 $F = ma = 7500 \text{ N}$ 方向水平向左

对飞机进行受力分析, 如右图所示.

对质心求矩, 则有

$$\begin{cases} N_B + N_R = G \\ N_B \times 12.6 = N_R \times 2.4 \end{cases}$$

$$\Rightarrow N_B = 2 \times 10^5 \text{ N}$$



9.12. 解: 如右图所示, 对杆OA进行受力分析.

惯性力作用在距O点 $\frac{2}{3}L$ 处

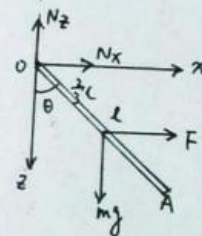
$$F = \frac{1}{2} L \sin \theta m \omega^2$$

对O点取矩

$$mg \cdot \frac{L}{2} \sin \theta = F \cdot \frac{2}{3} L \cos \theta$$

$$\Rightarrow \cos \theta = \frac{3g}{2\omega^2 L} \Rightarrow \theta = \arccos \frac{3g}{2\omega^2 L}$$

$$\text{当 } \theta = 0 \text{ 时, } \omega \text{ 最小, 为 } \omega_{\min} = \sqrt{\frac{3g}{2L}}$$



9.17. 解: 由平面运动刚体运动上惯性力矩可知.

$$\begin{cases} ma_x = T \sin 33^\circ \\ ma_y = T \cos 33^\circ - G \\ a = \sqrt{a_x^2 + a_y^2} \end{cases} \Rightarrow \begin{cases} a_x = 7.26 \text{ m/s}^2 \\ a_y = 1.38 \text{ m/s}^2 \end{cases}$$

$$-J_c \varepsilon + 3T \sin 33^\circ = 0 \Rightarrow \varepsilon = \frac{628}{J_c} \text{ rad/s}^2$$

9.16 对Q分析: $M_Q g = T + M_Q a_Q$

设圆柱中心加速度为 a_C

则顶端加速度 $a_1 = a_C + r \cdot \varepsilon$

$$a_1 = a_Q$$

对C分析: $T = f + m_C \cdot a_C$

$$(T + f) \cdot r = J_C \cdot \varepsilon$$

$$J_C = \frac{1}{2} m_C r^2$$

联立得 $\varepsilon = \frac{m_Q g}{(\frac{3}{4} m_C + 2 m_Q) r}$
 $= \frac{2g}{7r}$

9.18 问: $a_{\text{平动}} = \frac{F - f}{m}$
 $= \frac{11}{3} \text{ m/s}^2$

对A点取矩

$$I \cdot \varepsilon = (\frac{1}{3} m L^2) \cdot \varepsilon = F \cdot a_{9m}$$

$$\Rightarrow \varepsilon = \frac{25}{18}$$

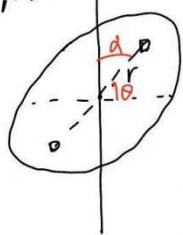
$$a_C = a + \varepsilon \cdot L$$

$$= \frac{11}{3} + \frac{25}{18} \times 3.6$$

$$= \frac{11}{3} + 5$$

$$= \frac{26}{3} \text{ m/s}^2$$

9.22



在圆盘上取两对
称小区域

$$dm = r d\theta \cdot dr \cdot \rho$$

$$\rho = \frac{m}{\pi R^2}$$

$$dF = dm \cdot \omega^2 r \cos \alpha$$

$$\begin{aligned} dM &= r \times dF = r \cdot dm \cdot \omega^2 r \cos \alpha \cdot \sin \alpha \\ &= r^3 \omega^2 \cos \alpha \cdot \sin \alpha \cdot \rho d\theta dr \end{aligned}$$

$$\begin{aligned} \therefore M &= \int_0^\pi \int_0^R \rho \cdot r^3 \cdot \omega^2 \frac{1}{2} \sin 2\alpha d\theta dr \\ &= \frac{1}{8} m \omega^2 R^2 \sin 2\alpha \end{aligned}$$

$$\begin{aligned} \because N_A &= N_B \\ \begin{cases} N_A \cdot a + N_B \cdot b = M \\ N_A = N_B \end{cases} &\Rightarrow N_A = N_B = \frac{m \omega^2 R^2 \cdot \sin 2\alpha}{8(a+b)} \end{aligned}$$

9.24. 解: 由定义可求出该薄板绕AC转动的转动惯量

$$J = \frac{m a^2 b^2}{6(a^2 + b^2)}$$

$$\text{则小惯性矩 } L = J \varepsilon = \frac{m a^2 b^2 \varepsilon}{6(a^2 + b^2)}$$

该力矩与 m 同向, 则有

$$M = J \varepsilon$$

$$\Rightarrow \varepsilon = \frac{6M(a^2 + b^2)}{m a^2 b^2}$$

该薄板在A、C处受到 N_A 、 N_C 两个力, F_1 和 F_2 为两部分转动所产生的离心力, 这两组力应满足平衡条件

$$\text{薄板离心力 } F_1 \text{ 作用点距C点 } l_1 = \left(\frac{b^4}{2\sqrt{a^2 + b^2}} \right)^{\frac{1}{3}}$$

$$F_1 = F_2 = \frac{a b m \omega^2}{12\sqrt{a^2 + b^2}}$$

对O点取矩

$$F_1 \cdot 2 \left(l - \frac{\sqrt{a^2 + b^2}}{2} \right) = N_A \cdot \frac{\sqrt{a^2 + b^2}}{2} \cdot 2$$

$$\Rightarrow N_A = N_C = \frac{a b m \omega^2}{6(a^2 + b^2)} \left[\left(\frac{b^4}{2\sqrt{a^2 + b^2}} \right)^{\frac{1}{3}} - \frac{\sqrt{a^2 + b^2}}{2} \right]$$

